

Developing an Educational Neuroscience of Category Learning

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ABSTRACT— There is a disconnect between neuroscience research on concept learning and representation (focusing on categories of concrete objects), and concept learning challenges in science education (which concern systems of causal relationships among objects and events). Bridging this gap will both inform theories of science learning and expand our understanding about how the brain learns and represents knowledge. We examine both literatures, point to where they converge and diverge, and offer paths forward for collaboration between science educators and cognitive neuroscientists to the benefit of each. In cognitive neuroscience, we describe advances in neuroimaging analyses based on cognitive models that directly relate cognitive processes to neural activity. Adapting science education content to support these model-based analyses offers unique opportunities to answer open questions about the cognition behind successful science learning. Further, adapting educational materials for use in neuroimaging may create useful learning exercises that offer unique affordances complementing typical classroom activities.

Concepts provide a lens through which we make sense of the world, carving the constant flux of raw experience into simpler useful abstractions that guide our actions. Given their importance, the question of how we learn them has been a central focus in both basic cognitive neuroscience research (aiming to understand mechanisms) and applied education research (aiming to improve concrete learning outcomes). But despite a shared focus, their respective

literatures remain fairly autonomous. The goal of this article is to highlight the untapped potential for researchers in these fields to work together and lay out multiple possible ways to do so. Our thesis is that concept learning has the potential to be a paradigmatic example of the success of educational neuroscience where both theory and application can be mutually benefitted.

We see three primary benefits of this integrated approach. The first is that neuroscientific methods improve understanding of the cognitive and motivational processes underlying successes and failures of learning and education. Specifically, neuroimaging data can make contributions to theoretical debates in education that behavioral data alone could not. The second benefit is that by engaging with real educational materials and contexts, theories in neuroscience are challenged to explain forms of concept learning that occur in the real world and not just those convenient to laboratory experimentation. This means greater external validity for theories about the mechanisms of conceptual learning. These first two benefits are relatively uncontroversial, but still far from trivial. The third, perhaps most controversial, is that educational neuroscience research can make direct recommendations to educational practice.

We suggest concept learning has the potential for all three of these benefits. Learning to categorize, and to use categorical knowledge as the basis of induction, are key aspects of concept learning, and importantly research on these cognitive processes offers a common set of quantitative methods that have been extensively used in basic and applied research (see Figure 1 for information about different methods for researching categorization used in the research reviewed throughout). We focus on categorization in science education because science is the domain of education where categorization methods have been most successfully applied.

In this paper, we selectively review the categorization literatures, and we view as best suited to show the potential for an integrated educational neuroscience (although we are not suggesting other work could not also provide these benefits).

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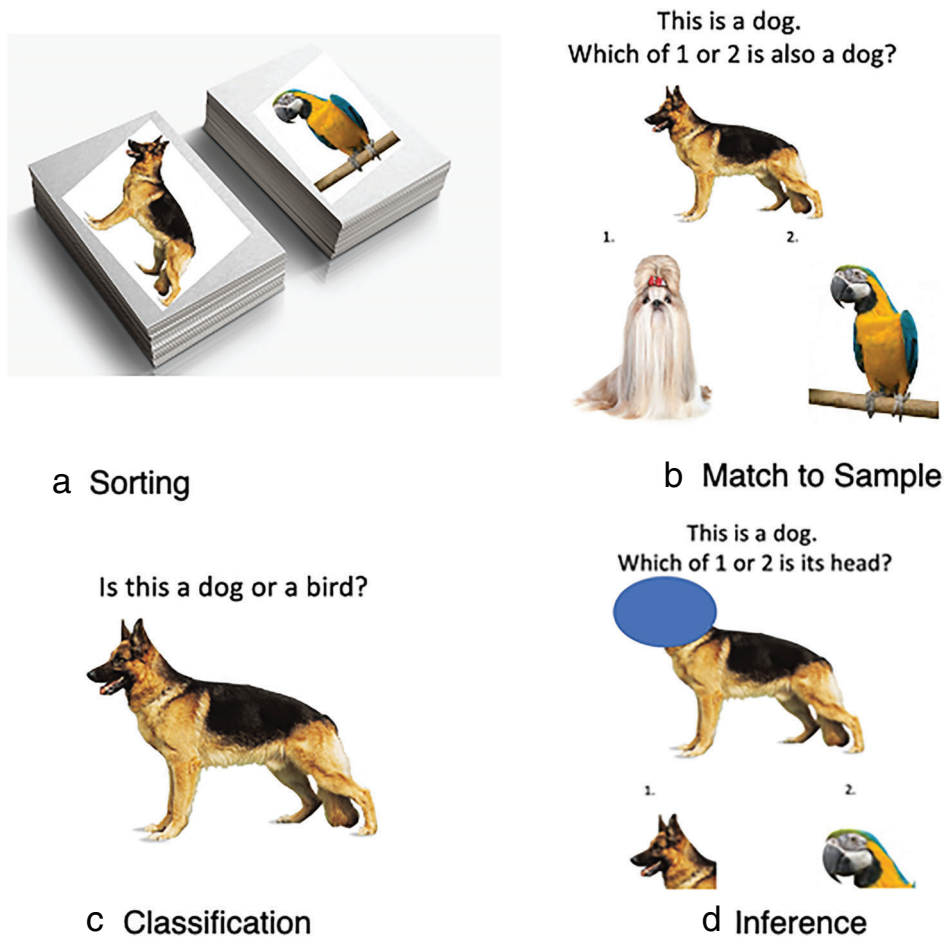


Fig 1. Four common categorization tasks. (a) Sorting: Participants can freely decide how to group stimuli. They can change their sorts multiple times throughout the process. There often no clear sequential trial structure or intermittent corrective feedback. (b) Match-to-sample: Participants are presented a target stimulus, and then indicate which of two other stimuli are in the same category. Trials are sequential; typically no corrective feedback. (c) Classification: Lone stimulus is presented, and participants guesses what category it belongs. Trials are sequential with corrective feedback in between trials. (d) Inference: Partial exemplar is presented, as well with its category. Then participant guesses which part completes the exemplar. Trials are sequential with corrective feedback in between trials.

The first section discusses behavioral research from education and cognitive psychology that shows how categorization changes as people acquire scientific knowledge, and what methods promote learning scientific categories. This establishes what can be done without any neuroscientific methods. The next section of the paper argues for the (potential) value added by neuroscience—that is, how the neuroscientific methods offer the above mentioned three benefits (to theories of learning from education; to theories of learning from neuroscience; and recommendations for educational practice). Last, we raise key open questions for moving the educational neuroscience of concept learning forward.

We also want to note, inspired by comments from an anonymous reviewer that we do not expect this paper to be particularly controversial. We are unaware of any factions within the field that is explicitly against applying basic

research on category learning to education; our goal is more to argue for the value of an approach that has not been given as much attention as other areas of cognitive research. Much of the field of educational neuroscience has focused on more general principles of learning, such as how sleep can help memory consolidation (e.g., Ribeiro & Stickgold, 2014), or relations between executive function and performance in school (e.g., Blair, Zelazo, & Greenberg, 2016), and less on the conceptual content of specific disciplines. When the content has been in focus, there is more research on math, literacy, and language learning than science (e.g., Kwok & Ansari, 2019). These lines of research are of course all quite valuable but are also limited in how they may be adapted by instructors to specific disciplines, such as in science education. As we discuss below, teaching specific disciplinary content raises unique learning challenges that cannot be

directly addressed by these other approaches, such as when scientific concepts conflict with our intuitive understanding of the world (see Shtulman, 2017 for review). We hope to show that there is a wealth of basic research from cognitive psychology and neuroscience that are quite relevant for discipline-specific education.

BEHAVIORAL RESEARCH ON CATEGORIZATION IN SCIENCE EDUCATION

Science Education Changes How Learners Categorize

Science education has gained direct insights from categorization research at least since Chi, Feltovich, and Glaser's (1981) classic paper on how novice and advanced physics students sort problems into categories. Novice students sorted primarily by the entities or *features* in the problems (such as pulleys, ramps, or whatever objects were asked to be reasoned about), while advanced students sorted by the underlying principles or causal *relations* relevant for solving the problems (such as the conservation of momentum). Focusing on relations, instead of features, is critical for abstracting the underlying concepts and applying them in new domains; one can make predictions about billiard balls or rocket ships—things with very different features, once one understands the relations underlying conservation of momentum.

Chi et al.'s (1981) work, and close to 40 years of subsequent research have shown that improvements in scientific problem solving over time reflect changes in how individuals conceive of, represent, and classify problems. A strong argument for focusing on categorization is that these conceptual changes are potentially more important for developing science expertise than other cognitive changes, such as changes to problem-solving procedural skills. This is because a key hurdle to accurate problem solving is knowing which procedure or principle to apply when (Goldwater & Schalk, 2016 for review). This is the hurdle that has to be overcome with accurate problem classification, and only after which refinements to procedural fluency are most useful.

There have been many similar demonstrations to Chi and colleagues using sorting tasks in the subsequent years. Rottman, Gentner, and Goldwater (2012) showed that students from an "integrated science program" could sort descriptions of phenomena from different fields based on common causal structure (such as grouping positive feedback systems in biology and economics together), while majors from individual social sciences (such as psychology and economics) sorted phenomena based on their field (e.g., putting all the biological examples into one category regardless of causal structure).

More directly relevant for performance in tertiary science courses, Stains and Talanquer (2008) showed that

introductory organic chemistry students categorized reactions by their products, such as water, while more advanced students categorized reactions by the reaction-mechanism, such as acid–base reactions. Similar to Rottman et al. (2012), these reaction-mechanisms reflect an abstract appreciation of causal structures and processes.

There are a growing number of papers examining the development of chemistry expertise using these kinds of sorting/categorization tasks (e.g., Galloway, Leung, & Flynn, 2018; Irby et al., 2016; Krieter, Julius, Tanner, Bush, & Scott, 2016). Again, a commonality across all this work is a change in conceptual organization. They all show a shift from focusing on some superficial aspect of a phenomenon, problem, or reaction, typically the intrinsic or perceptual features of the entities involved, toward a greater recognition of something deeper or more abstract, that concerns how all the entities interact or relate to one another. Dedre Gentner has called this "the relational shift" (Gentner & Rattermann, 1991). This shift of attention from the intrinsic features of entities (such as water from Stains & Talanquer, 2008) to the extrinsic relations between entities (such as the reaction mechanism) is associated with increased experience in a domain and is seen throughout the lifespan: in how children learn everyday concepts (see below), and in how adults become experts through higher education and professional experience (Gentner, 2010).

Organic chemistry is a particularly interesting field to examine a relational shift in students' conceptual organization (see Figure 2). As discussed in Galloway et al. (2018), traditionally, organic chemistry curriculum is organized to highlight functional groups, such as alcohols and carbonyls: a way of classifying molecules by their internal structure. Lectures or textbook chapters tend to focus on one functional group or another. Galloway and colleagues examine a different approach that was first implemented at the University of Ottawa in 2012, which emphasizes from the start of organic chemistry education "electron pushing" mechanisms. The same kinds of reaction mechanisms can apply to multiple kinds of functional groups. Although it is clear that students need to understand the links between molecules' internal structure and the reaction mechanisms that explain how multiple molecules causally interact, the historic emphasis on functional groups seems to have prevented many students from recognizing that these more general classes of mechanisms even exist or should be a target of learning. A typical anecdote from an organic chemistry student is that the class is an exercise in memorizing hundreds of reactions. Consistent with this, cognitive psychology research has shown memorization can be a default learning-strategy that students will resort to when they do not know to look for abstract principles (Goldwater, Don, Krusche, & Livesey, 2018; Little & McDaniel, 2015;

McDaniel, Cahill, Robbins, & Wiener, 2014). The curriculum from the University of Ottawa suggests that organic chemistry instruction can be reshaped to focus on reaction mechanisms, and that this can help develop a deeper conceptual understanding.

Linked with a conceptual reorganization reflecting a greater causal understanding, other research on developing science expertise has emphasized high-level perceptual changes (e.g., Goldstone, Landy, & Son, 2010; Kellman & Massey, 2013). This line of research has shown that understanding abstract causal processes in science is grounded in both static and dynamic visual representations (e.g., that simulate the causal processes) and that perception of these visual representations change with causal understanding. For example, geoscientists acquire new perceptual skills from the practice of inferring the causal processes that create geological features (Resnick & Shipley, 2013). Likewise, mathematical expertise fosters perceiving hierarchical structure in the written form of equations (Marghetis, Landy, & Goldstone, 2016). The point here is that a relational shift does not mean that somehow perceptual features are ignored, but they are perceived for how they illustrate or implement the more abstract causal relations. The deep relational structure is perceived in the features themselves (Chi & VanLehn, 2012).

This first section discussed changes in conceptual (and perceptual) organization with increased knowledge of science. Neuroscientific methods have rarely been applied to examining changes in conceptual organization (see below), but behavioral experiments both in the lab and in the classroom have made advances understanding this relational shift, and in showing what methods best support the process. We discuss this research next before arguing for how neuroscientific methods can take this work further.

Learning Relational Categories

The education research above shows a shift from novice categories defined by entities' intrinsic features ("feature-based categories") to expert categories defined by relations among entities ("relational categories"). This same distinction is relevant outside of formal education; our everyday conceptual repertoire is full of both feature-based and relational categories. Consider the concept *pet*. Although pets have typical features, such as four-legs, small, and furry, the concept is defined by the relationship between the pet and its owner. The more abstract and relational meaning was clearly highlighted when the concept was extended to the ironic fad of pet rocks in the 1970s (and see Gentner & Kurtz, 2005; Goldwater, Markman, & Stilwell, 2011; Markman & Stilwell, 2001 for thorough discussion).¹ We draw these connections between lay and formal concepts to better allow for application to formal learning settings.

These everyday relational categories show a similar developmental pattern to the relational categories from science education. In both cases, learners initially tend to miss the key underlying structure and instead focus on the superficial. Keil and Batterman (1984) showed that young children often initially interpret relational words, such as *island*, as having feature-based meanings, such as "place with beaches and palm trees", before learning the relational meanings, such as "land surrounded by water." Likewise, the same scaffolds that support learning nontechnical relational categories also support learning science concepts. For example, scaffolding learners to compare two examples that differ in their superficial features, but share underlying relational commonalities fosters categorization based on those underlying common relations (e.g., Gentner, Anggoro, & Klibanoff, 2011; Kurtz, Boukrina, & Gentner, 2013; Goldwater & Schalk, 2016 for review). This benefit of comparison for relational categorization has been shown in many studies, such as Goldwater, Bainbridge, and Murphy (2016), wherein naïve participants categorized novel computer-animated events, and Goldwater and Gentner (2015) where science novices learned to categorize phenomena from across domains of science that share causal structure (similar to how advanced science students from the integrated science program sorted in Rottman et al., 2012).

Those experiments just discussed had learners compare examples, and then later make categorization judgments. Kurtz and Honke (2019) went beyond this method and developed a category construction task as not just the measure of learning and conceptual organization, but as the learning experience itself. Although superficially, the task is quite similar to a sorting task. In this task, middle school science students were tasked to construct categories that grouped phenomena by their underlying scientific principle (such as evolution via natural selection). Later, their understanding of the scientific principle was assessed by whether they spontaneously applied their knowledge in a novel setting. Constructing a category out of multiple examples led to more spontaneous application of the underlying principle than just comparing a pair of examples.

Learning to Perceive Relational Structure With Corrective Feedback

To this point, the categorization research reviewed so far has used sorting tasks and match-to-sample tasks where the categorization judgments do not receive direct corrective feedback. Perhaps more common in experimental cognitive psychology are the tasks shown in Figures 1 and 3 that do present corrective feedback. By presenting stimuli, asking for rapid judgment, and providing corrective feedback, these tasks allow for learning trajectories over hundreds of trials to be examined in a single experimental session. These methods

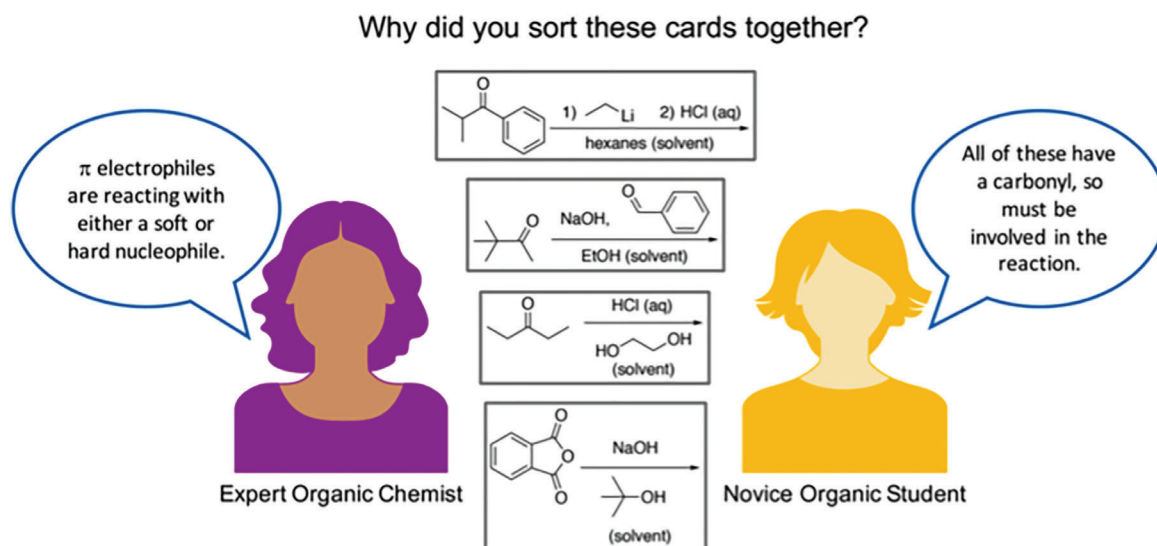


Fig 2. Image from Galloway et al. (2018) contrasting a novice sort based on the functional group carbonyl, and the expert sort based on reaction mechanism.

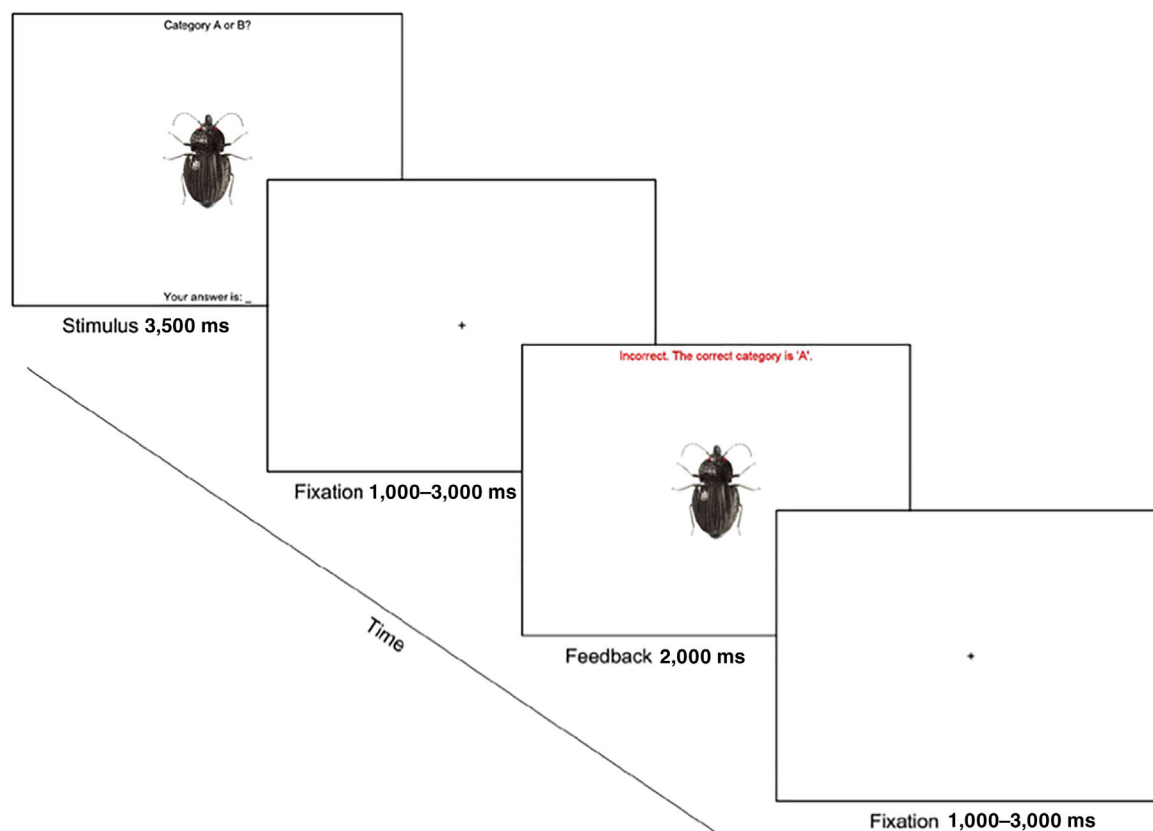


Fig 3. Figure 5 from Paniukov and Davis (2018), showing the experimental procedure for the learning phase of an inductive classification task.

have also been applied to science education. One example is from Mestre, Docktor, Strand, and Ross (2011) who presented physics problems to students and had them indicate which among a few presented options (such as conservation of momentum) was the principle relevant for solving the problem.

Although Mestre et al. (2011) show this kind of method can be applied to learning principles, these rapid-feedback methods have been more frequently applied to developing the perceptual aspects of STEM expertise discussed earlier (e.g., Goldstone et al., 2010). Kellman and Massey (2013) used similar methods to teach the mappings between functional expressions and graphs (where graphs are presented and learners select among possible equations), and how to read EKGs (wherein different patterns are presented and learners choose among possible irregularities).

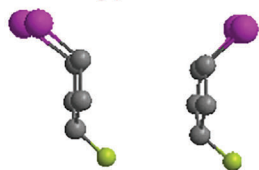
These tasks where learners are presented repeated exemplars, make judgments, and receive corrective feedback in a relatively rapid time scale have the potential to not just measure concept learning in science, but directly aid learning. Kellman and Massey (2013) showed that their function-to-graph learning methods greatly improved student performance, even in comparison to typical problem-solving practice. We follow their lead to make clear that these kinds of tasks are not meant to replace typical lectures, reading textbooks, nor problem-solving practice, but can offer a useful complementary form of instruction. Specifically, these tasks train rapid higher-level recognition skills. Kellman and Massey (2013) have argued that higher-level perceptual expertise critically supports deliberate reasoning because the right kind of information needs to be readily extracted from exemplars to support classification and then further reasoning. These rapid recognition skills are often what are lacking when students only learn from traditional instruction, and this greatly hampers their problem solving.

Tan (2016)² extended this line of work on perceptual and relational learning into organic chemistry, where

introductory students often struggle to identify key aspects of molecular structure. This work examined isomers, which are pairs of molecules that have the same chemical formula, for example $C_6H_{12}O_6$, but have different arrangements of atoms and thus different properties. There are many different kinds of isomers, defined by different kinds of contrasting relations between the pair of molecules, and students need to be able to readily classify a pair of molecules as the appropriate kind. The study focused on stereoisomers, which are defined by their spatial three-dimensional relationships. The study specifically contrasted two categories of stereoisomers that showed different kinds of spatial relational differences between the molecules in the pair.³ The molecules were represented in “ball and stick” diagrams (see Figure 4), without any atomic labels, such as “C” or “H.” The goal was to rid the task of potentially distracting technical language, so learners could focus on the spatial relationships. In a modified inductive classification task, naïve participants (i.e., not chemistry students), successfully learned to discriminate between the two categories of stereoisomers (they were not perfect, but better than randomly guessing). This study showed that people with no chemistry experience could learn to categorize isomers using just general spatial-perceptual abilities. This finding offers promise that these kinds of exercises could be a supplement to typical chemistry instruction to train perceptual skills. In addition, the findings suggest that presenting just representations of molecular structure without atomic labels could be beneficial. Further research is needed however to see how perceptual learning is integrated with a deeper conceptual understanding.

Although the above studies examined key real-world science concepts, they did not directly establish conflicts between categorizing based on superficial features and categorizing based on structural relationships. To do this, Goldwater et al. (2018) examined relational category learning with naïve participants and completely artificial stimuli (in that they reflected no real-world categories) but designed the experiments to be a laboratory model of this key conflict

Is the following pair an E-Pair or D-Pair?



E-Pair

D-Pair

Fig 4. Example trial from Tan (2016).

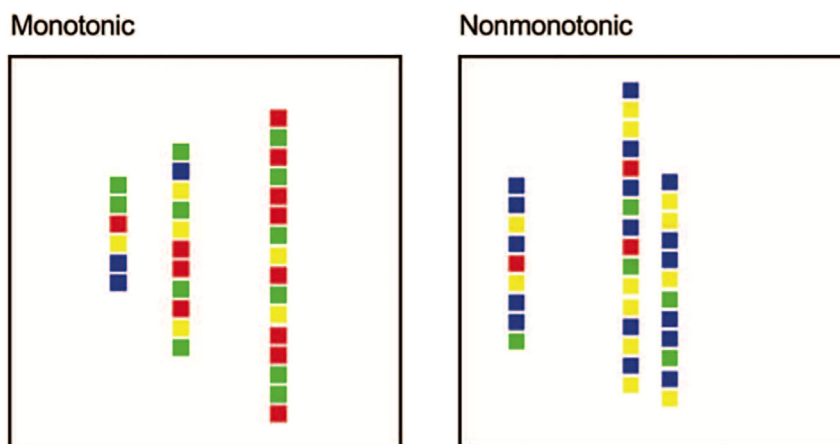


Fig 5. Stimuli from Goldwater et al. (2018). One category was defined by a monotonic change in line length from left to right, and the other category was defined by a nonmonotonic change in size from left to right. One category had more green and red squares, while the other had more blue and yellow squares.

between features and relations in developing expert-like concepts in science. The categories in this experiment were perfectly distinguishable based on spatial relations among objects, but there were correlated superficial features of the objects that probabilistically supported discrimination (see Figure 5). This structure mimics real relational categories, such as *predator* and *prey*, which have correlated but unessential features (e.g., predators tend to have frontward facing eyes, while prey animals tend to have eyes located more laterally on the sides of their heads).

Goldwater et al. (2018) produced multiple findings suggestive of how to shift focus of learners away from superficial features and toward categories defined by deeper relations. Here, we highlight one.⁴ The paper contrasted the classic inductive classification task with an “inference learning task” (Markman & Ross, 2003). Classification tasks present exemplars and ask to which category does the exemplar belong. Inference tasks present the category name and a partial exemplar. The learner then has to guess how to complete the exemplar such that it belongs to the appropriate category (see Figure 6). Both tasks have the same guess and corrective feedback trial sequence, and require learning about the categories to succeed, but evidence suggests they elicit learning which results in different representations of the categories. Past research showed that classification tasks focus learners on what discriminates the two categories, while inference tasks focus learners on the internal structure of categories. Goldwater et al. (2018) showed that inference tasks helped learners ignore the superficial features associated with the categories and learn the defining relational structure. This finding makes sense because the best way to predict a missing part of an exemplar is to understand the typical relations among the parts. A learner can conceive of how a missing part would fit in with the rest of the exemplar if they

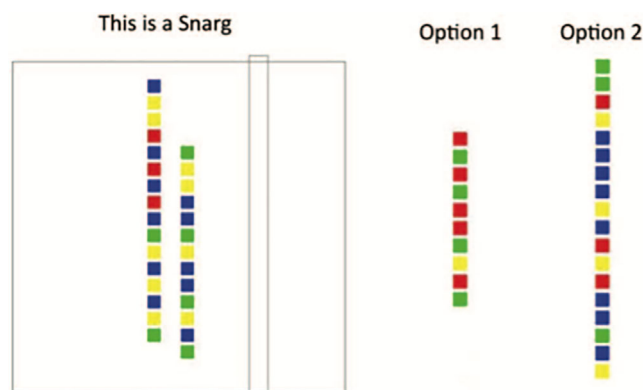


Fig 6. An example inference-learning trial where an incomplete exemplar is presented and two ways to complete the exemplar are offered as response choices.

understand how the parts relate in each category. This finding suggests that adapting inference learning tasks to science categories could support building expert-like conceptions, such as categories of organic chemical reactions based in electron-pushing mechanisms in Galloway et al., 2018 (see more below).

Learning to Change How to Categorize in the Lab

There is one more line of laboratory categorization tasks to discuss here because they examine when learners need to change how their categories are organized. These papers do not specifically examine a shift from a feature-based organization to a relational organization. Instead they focus on how learners change how they carve up a conceptual feature space; still the change process has some key commonalities to conceptual change processes in science.

Investigations of “knowledge restructuring” (e.g., Sewell & Lewandowsky, 2011) show how categorical knowledge is often partitioned into multiple component rules concerning different regions of conceptual space, and how people when prompted to and in the face of corrective feedback, can reorganize how these components relate (Kalish, Lewandowsky, & Davies, 2005). This kind of conceptual reorganization is similar to how Disessa and colleagues (e.g., Smith III, Disessa, & Roschelle, 1994) characterize shifts between novice and expert problem solving, where experts do not have entirely new knowledge, but have reorganized existing knowledge. Work on knowledge restructuring and science education suggest that conceptual representations can change without necessarily becoming more relational or abstract, but rather more efficiently partitioned, and re-coordinated on the fly.

Likewise, Hélié, Shamloo, and Ell (2020) examined how people learn new categorization schemes for stimuli they have previously learned to categorize. This work shows how the second learning pattern is dependent on the first. This pattern is similar to conceptual change research in science education, where learners do not simply learn new conceptions in formal education, and abandon their naïve conceptions about the natural world, but rather integrate formal concepts with their preconceptions (Vosniadou, 2012).

Summary

In this first part of the paper, we discussed the behavioral research that outlined how categorization changes with knowledge acquisition in science, and what category learning methods may best foster these changes. Next, we turn the benefits of adding neuroscientific methods to this program of research.

THE BENEFITS OF AN EDUCATIONAL NEUROSCIENCE OF CONCEPT LEARNING

The First Benefit: Improved Measurement of Cognitive Processes and Representations

The first benefit we listed above is how neuroimaging can measure cognitive processes in ways that go beyond what can be determined from behavior alone by providing a richer and dimensionally expanded set of correlates from which to infer the latent (internal) cognitive processes people are engaging to classify a given example as a member of a category. To show this, we need to first consider in more detail the inductive classification paradigm (see Figure 3) and discuss formal theories of categorization developed in the context of basic behavioral research.

Classification learning requires experimental participants to induce a way to classify novel stimuli into artificial categories (i.e., developed for the sake of the experiment). These

experiments typically present novel exemplars one at a time from one of two possible novel categories; the participants indicate to which category they think the exemplar belongs, and received corrective feedback (i.e., whether they were right or wrong). At first the participants simply guess, but over several trials, eventually build an understanding of what distinguishes the two categories from each other. Typically, this “learning phase” repeats a limited set of exemplars until an accuracy-criterion is met, such as reaching a certain percentage correct or number of trials without an error. The learning phase is followed by a generalization or “test” phase where participants see novel exemplars that were not presented in the learning phase. The participants classify these novel exemplars, but now no feedback is given (to limit further learning). The generalization phase is critical for understanding the category representations participants formed during the learning phase.

Typically, the goal of basic cognitive research on categorization is to understand how categories are represented, and what processes support category learning. There are many theories of category representation and learning, and typical experiments are designed to discriminate between these theories (rather than identify methods to improve learning outcomes). Categorization research has a long tradition of formally specifying theoretical accounts, which then make precise predictions for choice behavior in these paradigms. The most common models used in behavioral and neuroscientific categorization research assume that individual examples (objects to be categorized) are points in a multidimensional feature space. If learning about the natural categories of birds and mammals, dimensions may include things like whether the example has wings, fur, nurses their young, and so on. Categories are represented in these models in two primary ways: (1) as a sort of logical partitioning of the space in the form of rules (“if it has wings, it is a bird”; Ashby & Gott, 1988), or (2) as psychological distributions over points belonging to the categories (“choose which category the item is most similar to”). Distributional or similarity-based models can be represented in terms of simple parametric probability distributions (e.g., a normal distribution), as in prototype models (which captures an abstraction, generalizing across the observed exemplars), or may form a distribution based off of storing individual examples, as in exemplar models (Rosseel, 2002).

Model-Based Neuroimaging Analyses

Interestingly, despite the range of well-developed models of categorization, in many cases categorization choice does not constrain the underlying representational format (rules or similarity-based) sufficiently, just like it can be hard to know based on organic chemistry exam performance whether someone memorized individual examples

or learned a higher-level principle. Indeed, as discussed earlier, experts can devise complex rules for many categorization problems. Likewise, similarity-based models, like exemplar models, can learn category distributions of arbitrary complexity if given enough training data (Ashby & Alfonso-Reese, 1995). These issues have encouraged some researchers to look beyond behavior to neurobiological data for insight (Seger & Miller, 2010). For example, one popular neurobiological model of category learning COVIS (competition between verbal and implicit systems; Ashby, Alfonso-Reese, & Waldron, 1998) uses knowledge about the brain's learning systems to successfully account for how a number of category complexity and learning context manipulations will impact performance (Ashby & Maddox, 2011). Further, by using internal or "latent" processes from models to interrogate brain imaging data, one can begin to uncover unique brain signatures that are specific to predictions of rule-based or similarity-based models (or ones positing abstract relations, as we discuss later), giving unprecedented insight into how people are processing a problem (Davis & Goldwater, 2021).

Model-based neuroimaging differs from typical neuroimaging analysis that relies on the "subtraction" method. The subtraction method isolates a specific neurobiological process (e.g., "remembering") by subtracting brain activation during experimental trials in which a process is theoretically present (e.g., trials containing category members that are remembered during a later test) from a condition that is (supposedly) matched in all aspects not related to that process (e.g., trials containing category members that are later forgotten). Model-based imaging, on the other hand, looks for trial-by-trial signals that vary according to predictions from computational models. For example, an oft used model-based measure of how much participants may remember a category member is how similar the item is to previously stored category representations. Such model-based measures have accounted for activation in the medial temporal lobes, a brain region that is critical for explicit long-term memory, in a wide range of categorization tasks (Davis, Love, & Preston, 2012a, 2012b; Mack, Love, & Preston, 2016; O'Bryan, Walden, Serra, & Davis, 2018). Other common model-based measures include the error signals learning models use to update category representations and their connections (Davis et al., 2012b). Importantly, model-based imaging does not simply uncover processes underlying individual brain regions or structures but can inform how multiple brain regions work together. For example, Mack et al. (2016) showed how the prefrontal cortex, a brain region critical to how cognition helps us to achieve our goals (such as by filtering out irrelevant information), interacts with the hippocampus to change neural representations of exemplars to fit with the current goals of the categorizer.

Although model-based neuroimaging analysis frees categorization research from relying solely on behavioral data, participants' behavior often feeds into the model predictions as a constraint. The most frequent way that model-based predictions are linked to brain activation is by first fixing model parameters and predictions for latent processes (error-updating or remembering) based on the trial sequences participants saw and their trial-by-trial behavior. This leads to a unique trial-by-trial prediction for a given process that can then be incorporated into brain imaging analysis using typical linear regression procedures. However, more recent methods have begun jointly fitting both brain and behavior at the same time using Bayesian methods (Palestro et al., 2018).

So far, most model-based functional magnetic resonance imaging (fMRI) has focused on artificial, feature-based categories. To this end, the insights gained may account for some aspects of science learning and expertise, such as the development of automaticity for rules (Helie, Roeder, & Ashby, 2010), but do not directly test questions related to the relational-shift hypothesis that forms the inspiration for our review (because most models of categorization do not represent relations explicitly). Moving in this direction, Davis, Goldwater, and Giron (2017) were the first to contrast relational and feature-based category learning in an inductive classification task with fMRI. We first note that participants learned the feature-based categories and the relational categories using identical stimuli such that any differences between the two were not due to the perceptual characteristics of the stimuli. As with the studies discussed earlier, the fMRI data were analyzed with model-based methods (see the paper for model details).

Here, we highlight a few key findings from Davis et al. (2017). First, during the learning phase there was a key difference in activation patterns wherein learning relational categories elicited more activity in the rostral-lateral prefrontal cortex (rlPFC) than learning feature-based. The rlPFC is critical to a variety of higher-order cognitive processes, such as metacognition and long-term planning (e.g., Fleming, Huijgen, & Dolan, 2012; Koechlin, 2011) but most relevant it is critical to relational thinking (Vendetti & Bunge, 2014). The second important finding was that overall during the generalization phase when novel exemplars were presented, there were no differences in rlPFC between the two kinds of categories (unlike the learning phase). The third finding is perhaps most illuminating. During the generalization phase, the novel exemplars differed in how different the relations among the stimulus elements were from the learning exemplars. Critically, the greater the difference between the learning examples and relations of any given exemplar during generalization, the more rlPFC activation there was. This suggests that this brain region has a central role in extending the understanding of our

relational concepts to disparate exemplars.⁵ This relational extension process is importantly related to the chemistry example discussed earlier, wherein an understanding of an electron pushing mechanism needs to be extended to a novel reaction exemplar.

Other research on how people extend knowledge to accommodate new examples corroborates this key role for the rLPFC in higher-order relational reasoning. Anderson and Fincham (2014) use model-based analyses (using Anderson's cognitive architecture ACT-R; see paper for model details) of algebraic problem solving. The paper maps different aspects of problem solving to different brain regions, and critically showed that the rLPFC is activated when a known problem-solving procedure needs to be modified to solve a novel more complex problem.

Educational Neuroscience Benefit 1A Summary. Model-based analyses can reveal the roles of latent cognitive processes underlying concept learning, going beyond behavioral measures alone. However, up to this point, model-based analyses have not been applied to learning science concepts. The discussion in the last section is aimed to motivate that form of research.

Multivariate Pattern Analysis Techniques Better Constrain Theories of Conceptual Representation

The first two sections of the paper showed the evidence that learning science concepts entails a change in conceptual organization. In behavioral tasks, representational change is inferred through changes in sorting decisions, and how those sorting decisions relate to problem-solving abilities. However, another advantage of contemporary neuroscience methods is more fine-grained measures of conceptual representation.

Conceptual representations can be measured in neuroimaging data using multivariate-pattern analysis (MVPA; Davis & Poldrack, 2014; Iordan, Greene, Beck, & Fei-Fei, 2016). Here we are focusing on fMRI research (and *multivoxel pattern analyses*), MVPA analyses in other forms of neuroimaging (such as EEG and MEG; e.g., Cichy & Pantazis, 2017; Grootswagers, Wardle, & Carlson, 2017) offers similar advantages. MVPA capitalizes on the rich multidimensional nature of neuroimaging data to test questions about how task and concept-related information is coded in the relationships between elicited activation patterns for stimuli. Although such activation patterns carry many different signals related to task processing, they are currently the most direct way to test hypotheses about neural representations in human participants (Davis et al., 2014; Davis & Poldrack, 2014).⁶

In Davis and Poldrack (2014) participants first learned categories of novel bird drawings, and then rated how “typical”

each exemplar was of the category.⁷ During these ratings, they analyzed each exemplar's neural representation using MVPA. They showed that using MVPA during typicality judgments reveals the similarity in neural representations between exemplars and also explains the typicality ratings themselves. MVPA can also be used to identify what previously encoded exemplar is retrieved when considering how to generalize a decision to a novel example (Mack, Preston, & Love, 2013). Other MVPA studies have examined how weighting of relevant compared to irrelevant information emerges over the course of category acquisition (Mack et al., 2016; O'Bryan, Worthly, Livesey, & Davis, 2018), which can be used to test how the content of concept representations changes as a function of learning.

MVPA methods could readily be extended to novice and expert scientists to examine how neural representations change with experience. Similar to this idea, Cetron et al. (2017) used MVPA analyses to examine the effects of engineering education on object concepts. Images of, for example, bridges were shown to lay people and engineers, and MVPA revealed how engineers perceive these objects similarly too each other, but differently from naïve participants. The inference is that understanding the physical forces behind bridge design changes the neural representations of the bridge concept.

Educational Neuroscience Benefit 1B Summary. MVPA allows for more precise measures of conceptual representation than behavioral measures alone. Because representational change is at the heart of developing expertise in science, categorization tasks that show changes with experience such as Galloway et al. (2018) and Hélié et al. (2020) should be adapted for neuroimaging and MVPA. This will greatly advance our understanding of how science expertise develops.

Contributing to Theoretical Debates of How Conceptual Change Occurs

Although there is general agreement that conceptual representational change is a key part to science education, there is some theoretical disagreement to how this process occurs. Some theories of conceptual change argue that conceptual structures are fundamentally reshaped. Before formal education, people develop naïve or intuitive causal theories to explain the world, and these theories must be holistically restructured to accommodate formal scientific theories (Chi, 2009). Other accounts, as discussed earlier, focus on how new reasoners do not abandon previous conceptions as much as they re-coordinate previously held conceptual elements to be aligned with formal education (Smith III et al., 1994). MVPA analyses could contribute to this theoretical debate by showing the precise degree

of overlap between conceptual representations at different time points as students acquire greater scientific conceptual understanding. For example, by relating representational overlap, as measured with MVPA, to task performance, we can assess whether successful learning involves continual “reactivation” of previously learned information or if there are discrete time points during learning in which such information is largely abandoned, and whether individuals differ in their use of these strategies. These methods could also investigate how these MVPA patterns might be contextually mediated—with new learned patterns of activity in formal contexts but retained naïve patterns in certain everyday reasoning contexts.

A different theoretical debate about conceptual change is more directly relevant to instructional practice and could also be resolved by neuroimaging data. Some argue that inducing a cognitive conflict between novice and expert conceptions that supports learners’ awareness of the inadequacies of their prior conceptions is critical to effect conceptual change (Muller, Sharma, & Reimann, 2008; among others), while others have argued that cognitive conflict between conceptual schemes is not just unnecessary, but it is harmful to learning (e.g., Ramsburg & Ohlsson, 2016). Here model-based analyses can be helpful by allowing for precise formulation of conflict-based processes (e.g., Davis et al., 2012b) and looking at how the brain distributions differ from and relate to learning processes. For example, classic theories of conflict monitoring focus on anterior cingulate cortex (ACC) and dorsomedial prefrontal cortex, which may be isolatable both spatially and temporally from the relational learning signals we expect to underlie rLPFC function. We can also examine connectivity between these regions, which can help to establish whether conflict is largely inhibitory or facilitatory (and thus provide support for one of these competing accounts).⁸

Educational Neuroscience Benefit 1C Summary. By measuring conceptual representations with MVPA, or cognitive processes more generally with model-based analysis, theoretical debates about how conceptual knowledge changes with education are more systematically informed and in a better place to resolve competing hypotheses regarding their latent cognitive underpinnings. But there remains a need to extend the kinds of judgments and learning tasks used in this research to the contexts and knowledge-domains specific to science education.

The Second Benefit: Challenging Neuroscientific Learning Theories

To review: The goal of the cognitive neuroscience of categorization (as with the majority of the behavioral research on categorization) is quite distinct from the goal of categorization in education research. The goal of the basic research is to

identify the cognitive and neural processes that explain how people learn and represent the categories of our informal/lay conceptual repertoire; it is not primary to identify how to improve learning. To achieve this goal, tasks are designed to discriminate between different theories of categorization, and to identify the component cognitive processes underlying categorization, all the while, experimentally controlling factors that are deemed extrinsic, and satisfying the various practical constraints of conducting research. Pursuing these research goals may not require the categories in the experiments, or structures of the learning task, to bare any obvious resemblance to categories or learning tasks in the real world, as long as they are cleverly designed to reveal something about the relevant cognitive and neural mechanisms. This approach has certainly been successful for advancing more general theoretical matters but, which should now be clear, leaves many unanswered questions about learning categories in specific domains like science.

Categorization research in science education consistently shows a pattern of conceptual change via a relational shift. Learners initially focus on the superficial, and then with more experience, their conceptions re-organize reflecting a deeper understanding. In chemistry for example, this deeper understanding often concerns molecular causal processes underlying chemical reactions. Neuroscience research uses category-learning experiments with unfamiliar stimuli examining how concepts are formed in the first place; not how one form of conceptual organization needs to be changed to a more sophisticated form. Further, the concepts examined in cognitive neuroscience are typically about the intrinsic features of individual objects—the kinds of concepts students initially form from their learning materials when they are novices.

Examining the change from novice to expert conceptions will push the formal models of categorization, and the accounts of their corresponding neural implementations. It is simply unknown to what degree the kinds of mechanisms identified in the learning of feature-based categories (Zeithamova et al., 2019) will be shown to play the same kind of roles in learning to categorize based on deeper relational principles. For example, it is a plausible prediction from Mack et al. (2016) that prefrontal-hippocampal connections will be particularly important in conceptual change processes but testing that prediction will certainly reveal further unanticipated patterns of results. Likewise, research on highly trained feature-based categories suggests a shift from prefrontal “rule-based” representations to cortical representations that are more automatically activated when an example of a category is observed (Helie et al., 2010), but it is unclear whether these results will hold with relational categories that form the basis for our review. It may be that these categories enjoy a similar cortical shift once people achieve sufficient expertise, or such relational memory structures

may be dependent on the hippocampus, as is found with some types of relational memory (Cohen et al., 1999). Expert relational recognition may even recruit regions originally thought to be dedicated to face perception (Bilalić, 2016; Bilalić, Grottenhaler, Nägele, & Lindig, 2016).

Further, relating the relational-shift-based expertise to cortical automaticity in feature-based learning raises an important question regarding whether there is any faster route to expertise than simple repetition: with feature-based categories cortical automaticity simply depends on large amounts of learning trials and cannot be by-passed with any amount of insight. Critically, when categorization is automatized via extended repetition, it can become more tied to specific stimulus features and motor responses. On the other hand, insight accelerates relational learning, and expert reasoning in science allows for the flexible transfer of known principles to novel contexts (Corral, Kurtz, & Jones, 2018; Goldwater & Schalk, 2016).

Whichever way these theoretical issues resolve, putting neuroscience in closer contact with real-world concept learning (such as in science) ensures more representative and ultimately more justifiably generalizable theory.

The Third (Potential) Benefit: Recommendations for Educational Practice

Ultimately, regardless of the sophistication of neuroscientific methods, it is the understanding of and ability to influence learning in a practical way that matters to educationalists. It is immaterial how precise or innovative a neural measurement may be if it does not have educational “cash-value.” As we have emphasized in this review, a behavioral foundation is necessary for systematic cognitive inquiry (see also: Krakauer, Ghazanfar, Gomez-Marin, MacIver, & Poeppel, 2017). Contributions from neuroscience need to be synthesized with and not just added on top of that already gleaned from behavioral evidence. Further, for neuroscience of science category learning to be in the position to make clear recommendations for educational practice, more work on the behavioral side is still needed and this needs to be done in conjunction with thought and experiment in neuroscience. Here, we build on the work already reviewed to suggest how to move in this direction and where a preliminary synthesis might lead for educational practice.

Again, there is a clear pattern from categorizing based on more superficial information to categorizing based on abstract relational principles that changes as students acquire knowledge. This pattern is consistently measured with sorting tasks. However, the relational principle-based categorization is typically *not* the direct target of instruction, rather categorization is used to assess if the principles have been learned by other means. To set up a neuroscientific learning task that for example, examines how category

representations are updated with corrective feedback, the basic behavioral task with these kinds of materials would have to be developed.

Extending the findings of Goldwater et al. (2018), classification and inference tasks could be developed for electron-pushing mechanism reaction categories. An inference learning task could be designed to present the state of molecules at one-point of time in a reaction and asks the participant to choose among two options for the state at another point in time, knowing the kind of reaction it is (and then given corrective feedback). Although it would seem natural to consistently show the start-state to the participant and have them predict the end-state, going in the opposite direction as well may increase conceptual flexibility (Kellman & Massey, 2013; McNeil & Alibali, 2005). This kind of task could ideally show how to change students’ sorting behavior to be based on electron-pushing mechanisms, though of course while still maintaining their knowledge of functional-groups (similar to the success in Goldwater & Gentner, 2015).⁹

Once these behavioral methods have been developed, neuroimaging data could give unique insights into how the conceptual representations change due to iterative corrective feedback. This insight could then lead to a better understanding of what sorts of strategies are most effective at developing expert-like representations in learners. These different candidate learning strategies could be explicitly compared and not only on their immediate behavioral outcomes (like in traditional educational studies) but also with respect to underlying patterns of neural activity that occur concurrently with the moment of learning. This might especially help with distinguishing how different learning designs result in knowledge that is more or less likely to support far-transfer (e.g., Jacobson et al., 2020), which is invisible at the time of learning in terms of behavioral data, but may be resolvable with concurrent neural data.

Granted, ahead of time, there is no guarantee that the neuroscientific data would provide recommendations for practice that go beyond those already derived from behavioral data. But we can be certain of benefits 1 and 2: collecting neuroscientific data along with the behavioral data will inform both educational and neuroscientific theories. Until sufficient research is conducted along these lines we cannot be certain of benefit 3. However, the incentive to search for these neuroscientific insights are increased because to develop the methods to gain the theoretical benefits 1 and 2, and maybe 3, useful instructional tools will be developed and refined.

Adapting course materials for neuroscientific experimentation will along the way help to create effective learning activities for the course itself. For example, Kellman and Massey (2013) created math learning exercises that were at once appropriate for event-based

neuroscience research *and* improved learning more so than some typical classroom activities. In adapting organic chemistry learning tasks for fMRI to examine differences between focusing on functional groups or electron-pushing mechanisms, these tasks could help the goal of the University of Ottawa approach to teach a focus on these mechanisms. We are not trying to suggest it would be easy or fast to design effective category learning activities for real educational environments but given the success of research such as Kurtz and Honke (2019) and Mestre et al. (2011), it is potentially a worthwhile investment of time for both researchers and educators.

Summary of the Benefits of Educational Neuroscience

Behavioral research shows a clear pattern of conceptual reorganization as students acquire science knowledge. The first benefit of neuroscience is that model-based analyses can better reveal the underlying cognitive processes that support this conceptual reorganization, MVPA can show how conceptual representations change with knowledge acquisition on a fine-grain level, and these methods together can inform theoretical debates about the nature of the conceptual change process, such as the role of cognitive conflict. The second benefit is that neuroscientific theories will be challenged to account for learning real world complex materials that may conflict with learner's prior knowledge, rather than completely unfamiliar simple perceptual categories, ultimately improving the external validity (generalizability) of its findings. The third benefit is that adapting course materials for neuroscientific research may actually create useful learning exercises to complement typical instruction, and neuroscientific measurement may further inform how best to design instructional tools.

CONCLUSION

Our aim of this paper was to describe one path for research connecting cognitive neuroscience and science education, but in doing so, to motivate educational neuroscience of concept learning more broadly. The kinds of learning activities proposed near the end of the paper that could serve as methods for neuroscience research, and perhaps even used in classrooms would require tight collaboration between scientists and science educators (Coburn & Penuel, 2016, and e.g., Hilton et al., 2021). The authors of this paper could certainly not design the organic chemistry learning experiment described above because we have virtually no knowledge of organic chemistry. We hope our discussion motivates cognitive neuroscience researchers and science educators to work together to the benefit of both.

ENDNOTES

- ¹ A further connection with STEM education is that even with feature-based categories, people often represent rich causal relations among the features. For example, we do not just represent *guitar* as a list of features, such as *has strings*, *has a hollow wooden body*, and *makes music*, but we represent how those first two features are essential for producing the third (e.g., Sloman, Love, & Ahn, 1998).
- ² This study was an undergraduate honors thesis supervised by the first author of the current manuscript. The thesis is unpublished, so if any readers are interested in more information please contact the first author.
- ³ The two kinds of stereoisomers are enantiomers, which are chiral, meaning they are nonsuperimposable mirror images of each other, and diastereomers, which are not mirror images.
- ⁴ Another set of findings from this paper relate to specific learning sequences. These experiments compared the benefits of blocking sequences of exemplars from the same category versus interleaving exemplars from different categories. This is perhaps the area of category learning research where application to education has been discussed the most (see e.g., Bjork, Dunlosky, & Kornell, 2013; Carvalho, Braithwaite, de Leeuw, Motz, & Goldstone, 2016; Goldwater & Schalk, 2016). For brevity and because the discussion is readily available elsewhere, we will not describe this area of research in the current paper.
- ⁵ Paniukov and Davis (2018) suggests that rLPFC may not reflect an extending a category representation, but evaluating whether a learned categorization-rule can be applied to a novel example. The more disparate the example, the more evaluation needed (and hence greater rLPFC activation).
- ⁶ Here, we use "MVPA" to refer to a cluster of related analytic methods that all focus on multivoxel analyses, such as representational similarity analysis (RSA). In RSA, voxels are represented as points in space, and representational similarity between category exemplars (for example), would be calculated by the distance between the points of activated voxels. MVPA analyses can go further, and train classifiers based on patterns of activated voxels, for example, to show how exemplars of one category reliably elicit a distinct activation pattern compared to exemplars from another category.
- ⁷ One of the universal findings in categorization research is that categories are seen to have "graded membership." That is, there are more or less typical or "good" members of a category. For example, a crow is seen as a more typical, or better example of a bird than a penguin, despite in a technical biological sense, there are no better or worse

species of bird in the Aves class. People even see categories as formal as *odd number* has having graded membership; for example, 7 is a better odd number than 25 (Armstrong, Gleitman, & Gleitman, 1983).

- ⁸ There is work examining how people need to inhibit prior misconceptions to answer questions with scientific knowledge, but less on the process of conceptual change itself (e.g., Foisy, Potvin, Riopel, & Masson, 2015).
- ⁹ We thank an anonymous reviewer for asking whether it is actually advantageous to accelerate the relational shift, or rather is it necessary for learners to have an extended phase focusing on the more featural aspects. In many contexts, such as in organic chemistry, the more featural aspect, such as the functional group, is still important information, and is directly relevant to how the electron-pushing mechanisms work, so students need this knowledge as well. However, Goldwater and Gentner (2015) showed with causal system categories, that learning to categorize by causal system did not detract from the ability to categorize by the default strategy that learners started with. When asked, they could revert back to the default. The problem is that many learners will never grasp the more abstract relational principles without further instruction; they need less guidance to learn the more feature-based information. That said, any applied use of these methods should be careful that learners are not simply replacing one subset of important information with another subset of important information.

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